

Codebook: CFA-WM working memory strategies data (v1/v2/v3), participant- and stimulus-level exports

Data from the CFA-WM working memory task (word/orientation/colour recall under a parity-judgement distractor), pooled across three task versions (v1 original, v2 added a parity-error penalty, v3 added recall-order randomisation). Two datasets are described here: `all_data_surveys` (one row per participant: demographics, questionnaire scores, and raw questionnaire/strategy items) and `all_data_full` (one row per stimulus: trial structure, targets, responses, timing, and per-trial scores; no questionnaire item detail, see `all_data_surveys` for that).

Sources

- CFA-WM v1/v2/v3, Mael Delem, EMC Lab, Universite Lumiere Lyon 2

Instrument references and copyright notice

VVIQ, OSIVQ, and NIEQ item wording is copyrighted by the original scale authors and is not reproduced here. Item columns below are identified by their official item number/dimension; consult the source publications above for exact wording.

- VVIQ item text: Marks, D. F. (1973). Visual imagery differences in the recall of pictures. *British Journal of Psychology*, 64(1), 17-24. <https://doi.org/10.1111/j.2044-8295.1973.tb01322.x>
- OSIVQ item text: Blazhenkova, O., & Kozhevnikov, M. (2009). The new object-spatial-verbal cognitive style model: Theory and measurement. *Applied Cognitive Psychology*, 23(5), 638-663. <https://doi.org/10.1002/acp.1473>
- NIEQ item text: Heavey, C. L., Moynihan, S. A., Brouwers, V. P., Lapping-Carr, L., Krumm, A. E., Kelsey, J. M., Turner, D. K., & Hurlburt, R. T. (2019). Measuring the frequency of inner-experience characteristics by self-report: The Nevada Inner Experience Questionnaire. *Frontiers in Psychology*, 9, 2615. <https://doi.org/10.3389/fpsyg.2018.02615>

Dataset 1: `all_data_surveys` (one row per participant)

Core variables

Variable	Description	Type	Range	Levels / notes
<code>id</code>	Participant identifier, unique within a version.	string		
<code>version</code>	CFA-WM task version.	string		<code>v1</code> = Original task version; <code>v2</code> = Added a parity-error penalty to recall scoring; <code>v3</code> = Added recall-order randomisation; independent front-end infrastructure
<code>language</code>	UI language of the experiment in the browser.	string		<code>en</code> = English; <code>fr</code> = French

Variable	Description	Type	Range	Levels / notes
age	Participant age in years, self-reported.	integer	18-71 years	
gender	Gender, as freely typed by the participant.	string		Free text, not a controlled vocabulary; values include both short codes (“f”, “m”) and longer self-descriptions in French and English (e.g. “Non-binaire”, “female/agender”). Left as-is rather than collapsed into fixed categories, since collapsing would lose information some participants specifically chose to add. No missing values in the current data.
vviq_total_score	Total score on the Vividness of Visual Imagery Questionnaire (VVIQ; Marks, 1973), summed across all 16 items (see vviq_q01-vviq_q16 below). Lower scores indicate weaker/absent visual imagery.	integer	16-80	Missing for 2 participants (of 116).
vviq_group_2	Categorical VVIQ group with 2 levels, based on standard VVIQ cutoff.	string		aphantasia = VVIQ ≤ 32 (aphantasia and hypophantasia collapsed); typical = VVIQ > 32 (typical and hyperphantasia collapsed)

Variable	Description	Type	Range	Levels / notes
vviq_group_4	Categorical VVIQ group with 4 levels, based on standard VVIQ cutoffs.	string		aphantasia = VVIQ = 16 (complete absence of visual imagery); hypophantasia = VVIQ 17-32 (reduced visual imagery); typical = VVIQ 33-74 (typical visual imagery); hyperphantasia = VVIQ 75-80 (extremely vivid visual imagery) No missing values.
nieq_mental_imagery	Nevada Inner Experience Questionnaire (NIEQ; Heavey et al., 2019) mental imagery dimension score: mean of the frequency and proportion items for this dimension (see nieq_freq_mental_imagery / nieq_prop_mental_imagery in the nested item columns).	float	0-100	No missing values.
nieq_inner_voice	NIEQ inner voice/speaking dimension score (mean of frequency and proportion items).	float	0-100	No missing values.
nieq_emotions	NIEQ feelings/emotions dimension score (mean of frequency and proportion items).	float	0-100	No missing values.
nieq_sensory_focus	NIEQ sensory awareness dimension score (mean of frequency and proportion items).	float	0-100	No missing values.

Variable	Description	Type	Range	Levels / notes
<code>nieq_unsymbolised</code>	NIEQ unsymbolized thinking dimension score (mean of frequency and proportion items).	float	0-100	No missing values.
<code>object_mean</code>	OSIVQ (Blazhenkova & Kozhevnikov, 2009) object-imagery subscale mean.	float	1-4.64	Missing for 2 participants (same 2 as <code>vviq_total_score</code>).
<code>spatial_mean</code>	OSIVQ spatial-imagery subscale mean.	float	1-5	Missing for 2 participants (same 2 as <code>vviq_total_score</code>).
<code>verbal_mean</code>	OSIVQ verbal cognitive-style subscale mean.	float	1-5	Missing for 2 participants (same 2 as <code>vviq_total_score</code>).
<code>total_score_word</code>	Participant's total word-recall score, summed across all trials.	float	6-70.5	
<code>total_score_angle</code>	Participant's total orientation-recall score, summed across all trials.	float	0-64.5	
<code>total_score_color</code>	Participant's total colour-recall score, summed across all trials.	float	5-64	
<code>total_score</code>	Participant's combined total recall score across all modalities and trials.	float	11.5-195.5	
<code>prognosis</code>	Self-reported aphantasia status/prognosis, where collected.	string		no = No; yes = Yes

Variable	Description	Type	Range	Levels / notes
neuro_trouble	Self-reported neurological condition, freely typed.	string		Free text (French), not a controlled vocabulary, e.g. “TDAH”, “TSA”, “depression suite a un burnout”. NA throughout for v3 (not asked in that version’s demographics form); collected for v1/v2 only.
treatment	Self-reported relevant treatment.	string		no = No; yes = Yes – NA throughout for v3 (not asked in that version’s demographics form); collected for v1/v2 only.

VVIQ item-level responses (vviq_q01-vviq_q16)

The VVIQ has no subscales; its 16 items are grouped into 4 scenes of 4 items each, each rated 1 (perfectly clear and vivid) to 5 (no image at all). Item wording is not reproduced here for copyright reasons – see Marks (1973).

Variable	Description	Type	Range	Notes
vviq_q01	VVIQ item 1 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q02	VVIQ item 2 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
vviq_q03	VVIQ item 3 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q04	VVIQ item 4 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q05	VVIQ item 5 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q06	VVIQ item 6 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q07	VVIQ item 7 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q08	VVIQ item 8 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
vviq_q09	VVIQ item 9 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q10	VVIQ item 10 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q11	VVIQ item 11 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q12	VVIQ item 12 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q13	VVIQ item 13 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
vviq_q14	VVIQ item 14 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q15	VVIQ item 15 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q16	VVIQ item 16 of 16, from the scene “a country scene with trees, mountains, and a lake” (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

OSIVQ item-level responses

34 items rated 1 (totally disagree) to 5 (totally agree), across three scales (Object, Spatial, Verbal). Item order in the raw front-end data differs between v1/v2 and v3; the columns here are named consistently across both after unnesting by name (not position). Item wording is not reproduced here for copyright reasons – see Blazhenkova & Kozhevnikov (2009).

Variable	Description	Type	Range	Notes
osivq_q01s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q02v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q04v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q05s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q06o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q07s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q08v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q09v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q11o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q12o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q13o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q14s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q16v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q17s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q18o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q20o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q23o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q26o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q27s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q29o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q30s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q31s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q32s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q33o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q34o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q35v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q37v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q39v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q40o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q41v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q42s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q43o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q44s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q45o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

OSIVQ subscale sums and item-completion counts

Variable	Description	Type	Range	Notes
object_score	OSIVQ object-imagery subscale sum (raw item sum; object_mean above is its mean).	integer	14-65	Missing for 2 participants overall.
spatial_score	OSIVQ spatial-imagery subscale sum (raw item sum; spatial_mean above is its mean).	integer	11-55	Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
verbal_score	OSIVQ verbal cognitive-style subscale sum (raw item sum; verbal_mean above is its mean).	integer	12-45	Missing for 2 participants overall.
o_count	Number of object-scale items answered (out of 11).	integer		Diagnostic/completion count, not a score.
s_count	Number of spatial-scale items answered (out of 11).	integer		Diagnostic/completion count, not a score.
v_count	Number of verbal-scale items answered (out of 12).	integer		Diagnostic/completion count, not a score.

NIEQ item-level responses

5 dimensions (mental imagery, inner voice/speaking, feelings/emotions, sensory awareness, unsymbolized thinking), each with 2 raw items (frequency, proportion), both 0-100 sliders. Item wording is not reproduced here for copyright reasons – see Heavey et al. (2019).

Variable	Description	Type	Range	Notes
nieq_freq_mental_imagery	NIEQ Mental imagery dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_mental_imagery	NIEQ Mental imagery dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.

Variable	Description	Type	Range	Notes
nieq_freq_inner_voice	NIEQ Inner voice/speaking dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_inner_voice	NIEQ Inner voice/speaking dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_freq_emotions	NIEQ Feelings/emotions dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_emotions	NIEQ Feelings/emotions dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.

Variable	Description	Type	Range	Notes
nieq_freq_sensory_NIEQ nieq_freq_sensory_NIEQ	NIEQ Sensory awareness dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_sensory_NIEQ nieq_prop_sensory_NIEQ	NIEQ Sensory awareness dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_freq_unsymbolized_NIEQ nieq_freq_unsymbolized_NIEQ	NIEQ Unsymbolized thinking dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_unsymbolized_NIEQ nieq_prop_unsymbolized_NIEQ	NIEQ Unsymbolized thinking dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.

Strategy report, free text (strats_cfa_q01-q03)

Free-text self-reported memorisation strategies, mostly in French, collected in v1/v2 only. Not a controlled vocabulary – no level list is given, as one wouldn't meaningfully summarise free text.

Variable	Description	Type	Range	Notes
strats_cfa_q01_colours_1	Free text (mostly French) self-reported strategy for remembering colours, first mention.	string		Free text, not a controlled vocabulary - collected in v1/v2 only. See strats_cfa_q04_scoring_strat_* below for v3's coded strategy report.
strats_cfa_q01_colours_2	Free text self-reported strategy for remembering colours, second mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q01_colours_3	Free text self-reported strategy for remembering colours, third mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q02_orientations_1	Free text self-reported strategy for remembering orientations, first mention.	string		Free text, not a controlled vocabulary - collected in v1/v2 only.
strats_cfa_q02_orientations_2	Free text self-reported strategy for remembering orientations, second mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q02_orientations_3	Free text self-reported strategy for remembering orientations, third mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q03_words_1	Free text self-reported strategy for remembering words, first mention.	string		Free text, not a controlled vocabulary - collected in v1/v2 only.

Variable	Description	Type	Range	Notes
<code>strats_cfa_q03_words2</code>	Free-text self-reported strategy for remembering words, second mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
<code>strats_cfa_q03_words3</code>	Free-text self-reported strategy for remembering words, third mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).

Strategy report, coded (`strats_cfa_q04_scoring_strat_*`)

Controlled vocabulary, collected primarily in v3 (`strats_cfa_q04_scoring_strat_1/_2`); v1/v2 has an equivalent field with a different naming convention (`_1_1/_2_1/_3_1`).

Variable	Description	Type	Range	Levels / notes
<code>strats_cfa_q04_scoring_strat_1</code>	Collected self-reported memorisation strategy, first choice.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v3 only (as <code>strats_cfa_q04_scoring_strat_1</code> ; the <code>_1_1</code> column below is a v1/v2 field with a different naming convention, not the same question re-asked).

Variable	Description	Type	Range	Levels / notes
strats_cfa_q04_score	Closest strat_2 self-reported memorisation strategy, second choice.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v3 only.

Variable	Description	Type	Range	Levels / notes
<code>strats_cfa_q04_score</code>	Closest strat_1_1 self-reported memorisation strategy, v1/v2's equivalent first-choice field.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v1/v2 only (NA for all v3 rows).

Variable	Description	Type	Range	Levels / notes
<code>strats_cfa_q04_score</code>	Collected self-reported memorisation strategy, v1/v2's equivalent second-choice field.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v1/v2 only (NA for all v3 rows). Sparsely populated even within v1/v2 (optional field).

Variable	Description	Type	Range	Levels / notes
strats_cfa_q04_score	Closest strat_3_1 self-reported memorisation strategy, v1/v2's equivalent third-choice field.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v1/v2 only; v3 has no third slot for this field (NA for all v3 rows - this is a real structural difference, not missing data). Rarely populated even within v1/v2: non-missing for only 7 of 88 v1 participants and 0 of 9 v2 participants.

Additional demographics

Variable	Description	Type	Range	Notes
country	Self-reported country of residence (free text/country code, not standardised).	string		
job	Self-reported occupation, using the ESEG occupational classification codes where applicable.	string		
education	Self-reported education level, using ISCED classification codes where applicable.	string		
field	Self-reported field of study or work, free text.	string		
where_from	Self-reported source of how the participant heard about the study.	string		
language_native	Participant's self-reported native language.	string		
language_usual	Participant's self-reported language of everyday use, where different from native language.	string		

Dataset 2: `all_data_full` (one row per stimulus)

`id`, `version`, `language`, `age`, `gender`, `vviq_total_score`, `vviq_group_2`, `vviq_group_4`, `nieq_mental_imagery`, `nieq_inner_voice`, `nieq_emotions`, `nieq_sensory_focus`, `nieq_unsymbolised`, `object_mean`, `spatial_mean`, and `verbal_mean` are the same participant-level columns described in `all_data_surveys` above, repeated on every stimulus row for this participant. See `all_data_surveys` for their descriptions, ranges, and missingness.

Variable	Description	Type	Range	Levels / notes
<code>expe_phase</code>	Task phase this stimulus belongs to.	string		<code>tutorial</code> = Single practice trial with feedback; <code>training</code> = Practice trials, no feedback; <code>expe_block_1</code> = Test block 1; <code>expe_block_2</code> = Test block 2; <code>expe_block_3</code> = Test block 3
<code>trial_number</code>	Trial index (1 tutorial + 3 training + 21 test trials per participant).	integer		
<code>response_order</code>	Order in which the three response modalities (word/angle/colour) were collected for this trial.	string		<code>word_angle_color</code> = Word, then angle, then colour; <code>word_color_angle</code> = Word, then colour, then angle; <code>angle_word_color</code> = Angle, then word, then colour; <code>angle_color_word</code> = Angle, then colour, then word; <code>color_word_angle</code> = Colour, then word, then angle; <code>color_angle_word</code> = Colour, then angle, then word – Fixed at <code>word_angle_color</code> in v1/v2 (always that order); randomised per trial in v3, so all 6 permutations appear.
<code>item_number</code>	Stimulus index within the trial (1-3: word, orientation, colour).	integer		
<code>target_word</code>	The to-be-remembered word for this stimulus.	string		

Variable	Description	Type	Range	Levels / notes
target_angle	The to-be-remembered rectangle orientation, in degrees.	integer		
target_color_angle	The to-be-remembered colour, as an angle on the colour wheel used for continuous colour report.	float		
target_color	The to-be-remembered colour, as its nearest named colour.	string		
response_word	Participant's recalled word.	string		
response_angle	Participant's recalled rectangle orientation, in degrees.	float		
response_color_angle	Participant's recalled colour, as an angle on the colour wheel.	integer		
response_color	Participant's recalled colour, as its nearest named colour.	string		
diff_word	Word recall error (string-distance based; 0 = exact match).	float		
diff_angle	Angular error between target and response orientation, in degrees.	float		
diff_color	Angular error between target and response colour on the colour wheel.	float		
rt_word	Response time for the word recall response.	integer		
rt_angle	Response time for the orientation recall response.	float		

Variable	Description	Type	Range	Levels / notes
rt_color	Response time for the colour recall response.	float		
score_word	Per-modality recall score for the word response, derived from diff_word.	float		
score_angle	Per-modality recall score for the orientation response, derived from diff_angle.	float		
score_color	Per-modality recall score for the colour response, derived from diff_color.	float		
trial_score	Combined recall score for this stimulus's trial, across all three modalities.	float		
parity_1_stim	First of the two parity-judgement distractor digits embedded in this trial.	integer		
parity_1_resp	Participant's parity judgement (odd/even) for parity_1_stim.	string		
parity_1_acc	Accuracy of the first parity judgement (1 = correct, 0 = incorrect).	integer		
parity_1_rt	Response time for the first parity judgement.	integer		
parity_2_stim	Second of the two parity-judgement distractor digits embedded in this trial.	integer		
parity_2_resp	Participant's parity judgement (odd/even) for parity_2_stim.	string		
parity_2_acc	Accuracy of the second parity judgement (1 = correct, 0 = incorrect).	integer		

Variable	Description	Type	Range	Levels / notes
parity_2_rt	Response time for the second parity judgement.	integer		
total_score_word	Participant's total word-recall score, summed across all trials.	float		
total_score_angle	Participant's total orientation-recall score, summed across all trials.	float		
total_score_color	Participant's total colour-recall score, summed across all trials.	float		
total_score	Participant's combined total recall score across all modalities and trials.	float		
prognosis	Self-reported aphantasia status/prognosis, where collected.	string		no = No; yes = Yes
neuro_trouble	Self-reported neurological condition, freely typed.	string		Free text (French), not a controlled vocabulary. NA throughout for v3.
treatment	Self-reported relevant treatment.	string		no = No; yes = Yes – NA throughout for v3.